

STUDIO 5000 ANALOG SCALING - FUNCTION BLOCK and STRUCTURED TEXT

FUNCTION BLOCK EXAMPLES

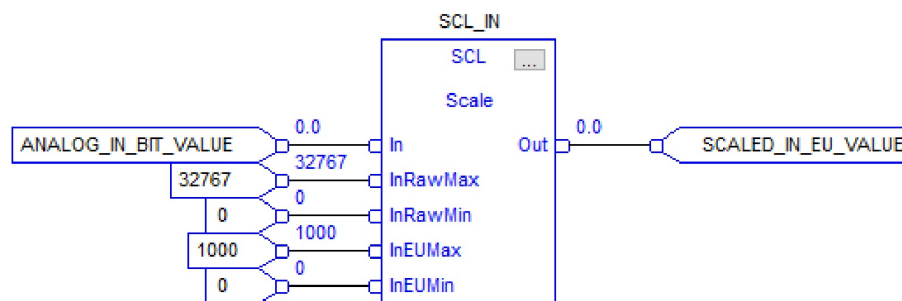
When the **SCL** instruction is used in the program the **SCALE Data Type** is defined with the numbered instance of the SCL instruction in the Parameters and Local Tags.

For an Add On Instruction, the numbered instance of the **SCL** instruction **Tag** with the **Data Type** of **SCALE** is define in the dialog Add-On Instruction Definition as a **Local Tags**.

Analog Input

Bit Value: 0 to 32767

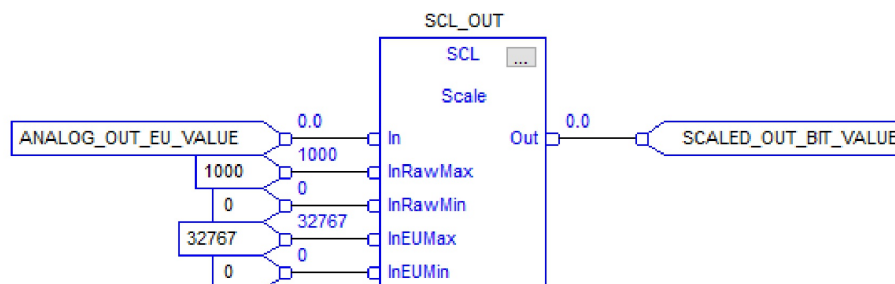
Engineering Units: 0 to 1000



Analog Output

Engineering Units: 0 to 1000

Bit Value: 0 to 32767

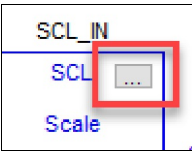


NOTICE: For the Analog Output the **Raw** and **EU** values are swapped.

FUNCTION BLOCK PARAMETERS

In the Function Block's Parameters, the Raw and EU "pins" may be unchecked and the values entered in the Parameter's field. **NOTE:** *The SCL instruction's Parameter Table is duplicated in the Tag Editor and the values could be changed for the assigned Tag, such as, by a MOV instruction.*

Instructions's Parameter button.



Properties - SCL_IN

Parameters Tag

	Vis	Name	Value	Type	Description
I	☐	EnableIn		1 BOOL	Enable Input. If False, th...
I	☒	In	0.0	REAL	The analog signal input to...
I	☐	InRawMax	32767.0	REAL	The maximum value attain...
I	☐	InRawMin	0.0	REAL	The minimum value attain...
I	☐	InEUMax	1000.0	REAL	The maximum scaled valu...
I	☐	InEUMin	0.0	REAL	The minimum scaled valu...
I	☐	Limiting	0	BOOL	Limiting selector. If TRUE...
O	☐	EnableOut		0 BOOL	Enable Output.
O	☒	Out	0.0	REAL	This is the output of the S...
O	☐	MaxAlarm		0 BOOL	The above maximum input...
O	☐	MinAlarm		0 BOOL	The below minimum input ...
O	☐	Status	16#0000_0000	DINT	Bit mapped status of the f...
O	☐	InstructFault		0 BOOL	Instruction generated a fault
O	☒	InRawRangeInv		0 BOOL	InRawMin <= InRawMax

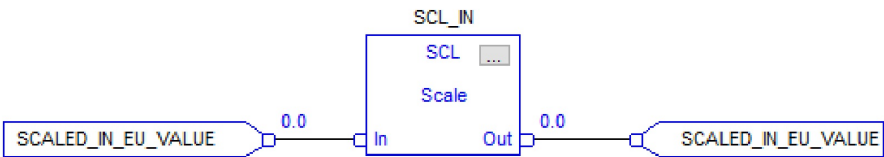
☐ Sort Parameters

Insert Instruction Defaults
Insert Factory Defaults
Save Instruction Defaults

Status: OK
Execution Order Number: 1
☐ Never display description in a routine

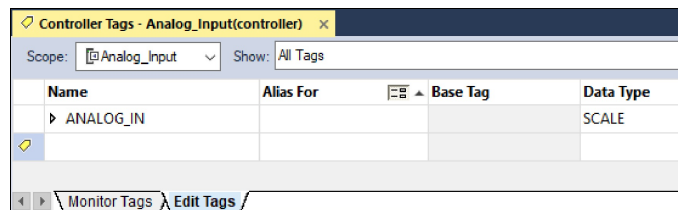
OK Cancel Apply Help

The SCL instruction with just the In and Out pins.



STRUCTURE TEXT EXAMPLES

A **Tag** for the **SCL** instruction using the **Data Type** of **SCALE** must be defined, as a Controller or Parameters and Local Tags.



Name	Alias For	Base Tag	Data Type
▶ ANALOG_IN			SCALE

For an Add On Instruction, for the **SCL** instruction the **Tag** and the **Data Type** of **SCALE** must be define in the dialog Add-On Instruction Definition as a **Local Tags**.

Analog Input

Bit Value: 0 to 32767

Engineering Units: 0 to 1000

```
//Analog Input to measure process value 0 rpm to 1000 rpm.  
//InRaw uses Bit Values 0 to 32767.  
//InEU uses Process Values 0 to 1000.
```

```
ANALOG_IN.InRawMin:=0;  
ANALOG_IN.InRawMax:=32767;  
ANALOG_IN.InEUMin:=0;  
ANALOG_IN.InEUMax:=1000;  
ANALOG_IN.In:=ANALOG_IN_BIT_VALUE;  
SCL(ANALOG_IN );  
SCALED_IN_EU_VALUE:=ANALOG_IN.Out;
```

Analog Output

Engineering Units: 0 to 1000

Bit Value: 0 to 32767

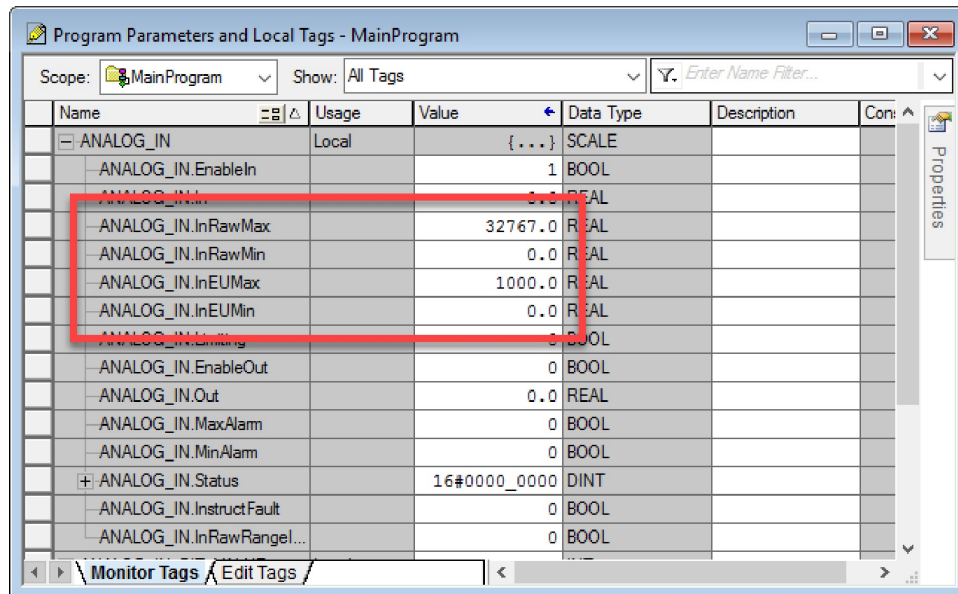
```
//Analog Output to control process value 0 rpm to 1000 rpm.  
//InRaw uses Process Values 0 to 1000.  
//InEU uses Bit Values 0 to 32767.  
//NOTE: The Bit and Process Values are swapped.
```

```
ANALOG_OUT.InRawMin:=0;  
ANALOG_OUT.InRawMax:=1000;  
ANALOG_OUT.InEUMin:=0;  
ANALOG_OUT.InEUMax:=32767;  
ANALOG_OUT.In:=ANALOG_OUT_EU_VALUE;  
SCL(ANALOG_OUT );  
SCALED_OUT_BIT_VALUE:=ANALOG_OUT.Out;
```

NOTICE: For the Analog Output the **Raw** and **EU** values are swapped.

STRUCTURE TEXT TAG EDITOR

The Structured Text's Tag in the Tag Editor Row and EU values may be entered in the Tag's field. **NOTE:** The SCL instruction could be changed for the assigned Tag, such as, by a MOV instruction.



NOTICE: The values are entered at the **Monitor Tags** tab.

The SCL instruction with the IN and OUT defined.

```
//Analog Input to measure process value 0 rpm to 1000 rpm.  
//InRaw uses Bit Values 0 to 32767.  
//InEU uses Process Values 0 to 1000.  
  
ANALOG_IN.In:=ANALOG_IN_BIT_VALUE;  
SCL(ANALOG_IN );  
SCALED_IN_EU_VALUE:=ANALOG_IN.Out;
```